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Poster: PAD – the Percussive Audio Database: Initial Concept and Design

Abstract

The organization and identification of sound files in sample libraries is a topic of concern in the fields of Music Information Retrieval and Timbre Classification. Of interest is the notion of labelling and searching for files using sonic and timbral criteria.

The author proposes classifying percussive sounds using the 'Percussive Audio Database' (PAD).

PAD is a multi-dimensional taxonomy specifically for drum and percussion libraries, em-ploying an Object-Relational model. It de-scribes sounds by a set of 34 classes, each one defining a specific spectral, temporal or timbral attribute. Examples include Spectral Density, Decay Rate and Attack 'Hardness'. 'Variables' such as beaters, excitation methods and morphology are also classified as potential modifi-ers of a given sound. PAD's object-relational model allows both hierarchical and nonhierar-chical relationships between data to be repre-sented. Classes are labelled with an alpha-graphical notation and visually displayed as a three dimensional matrix/flow chart, using hierarchical layers and distance measures to connect them.

Software is being designed that automatically analyses samples for spectro-temporal data, and permits user-input of various perceptual responses. Samples are then automatically classified with all 34 attributes. The user is able to search, filter and compare any classes between samples in a library.

Artificial Intelligence models are also explored, with a view to allowing PAD to 'learn' pat-terns of similarity and difference between dif-ferent samples in a collection. This in turn would further automate the classification proc-ess, and improve efficiencies in updating li-braries with new samples – especially shared libraries housed online and on local networks.